DAY-10

How Array work in Java ?

```
int [] arr = new int [5];
```

→ `System.out.println(arr[2]);`

Random Access.

Data Types :

- primitive
  - Integer
  - Float
  - Boolean
  - Char
- NON primitive
  - Arrays

```
int x = 4;
```

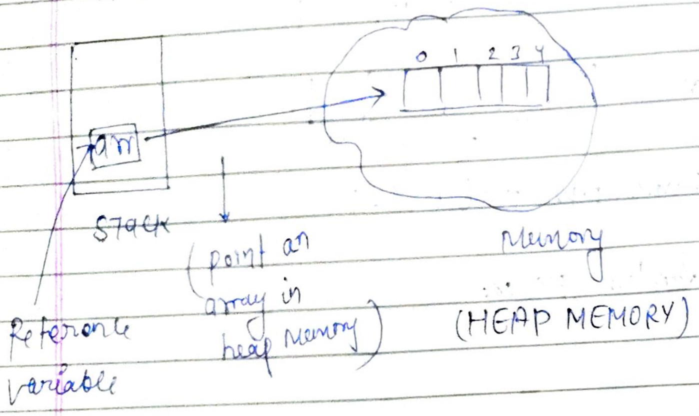
x $\boxed{4}$

→ mean 'x' directly hold the value 4.

Stack Memory

primitive Data type

int [] arr = new int [5];



arr - it does not directly holds the array
- It is a reference to an array

100 104 108 112 116 120 ← ~~start~~ address of Memory.

10	20	30	40	50

0 1 2 3

arr[0] = 10
arr[1] = 20
arr[2] = 30
arr[3] = 40
arr[4] = 50

System.out.println (arr[3]);

arr[3] = 100 + (4 x 3) = 112
arr[4] = 100 + (4 x 4) = 116

$$\boxed{arr[i] = \text{Base Address} + (\text{D.T size} \times i)}$$ #

(JVM use it)

int x = 4

int → 4 Bytes

x = 4;

4

100 104

→ arr - store starting address / Base Address of any array in heap

int [] arr = new int [5];

long - 8B char - 2B.
Float → 4B
Double → 6B.

Boolean - According to official java docs.

[T] [F]

(No fix size).

hotspot (oracle)
Open JDK)
↳ Boolean → 1B.

Q. Why not have fix size of Boolean.?

Boolean

T	F	
1	0	0/1

For better CPU optimize Boolean have 1B size

#

"CPU fetch data as Byte by Byte not bit by bit"

boolean [] arr = new boolean [5];

F	T	F		
0	1	2	3	4
100	101	102	103	104

105

arr [2] = 100 + (1 x 2) ⇒ $\boxed{102}$ ✓

→ arr =

0	1	2	3	4	— —	204	203

(Base Address) ← 100

arr [100]

└ 100 + (4 x 100) = 204..

Arrayindexoutofbound

```
if(index < 0 || index > arr.length) {
    throw Arrayoutofbound;
}
```

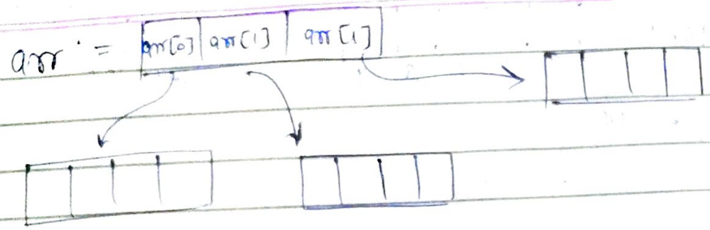
2-D Arrays :

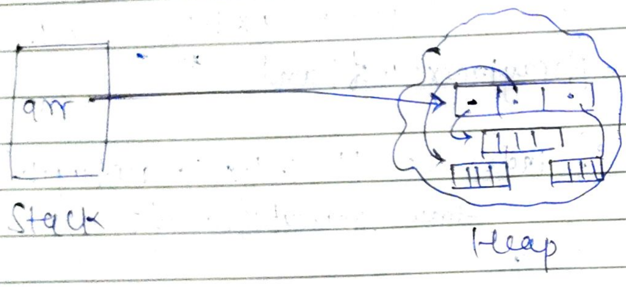
int [][] arr = new int [3][4]

	0	1	2	3
0	4	5	6	7
1	8	9	10	1
2	5	3	2	1

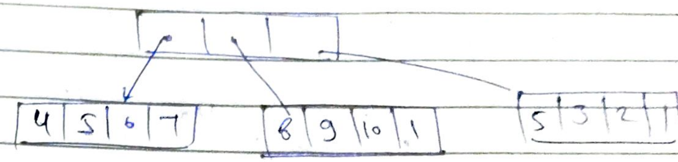
arr[1][2]

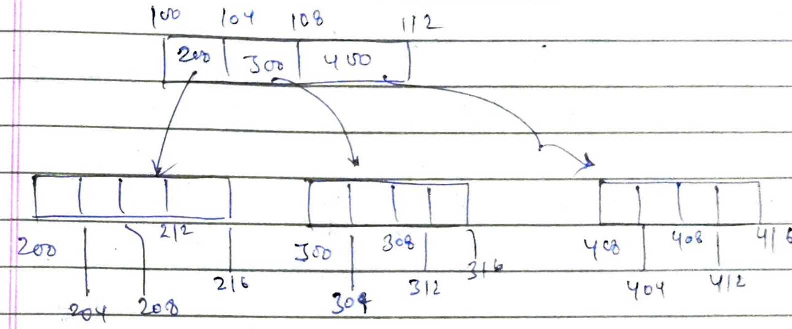
Q. How random accessl achieve ?

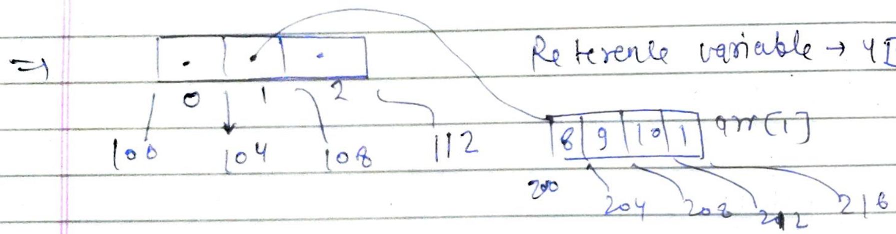




Memory view







Reference variable → 4B.

row = 100 + (4 x 1) = 104.

col = 200 + (4 x 2) ⇒ 208

$\boxed{arr[1][2] = 6}$ ans

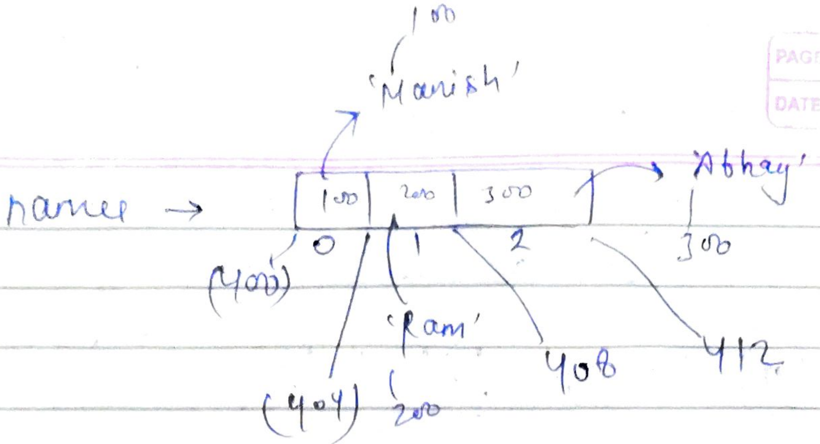
Strings:

String [] names = new String [3];

names → | 'Manish' | 'Ram' | 'Abhay' |
0 1 2

String → ?? (size)

→ names [1];



names[1]

$$\Rightarrow 400 + (4 \times 1)$$
$$= 404$$

** { think like char. Array }

{ Upto JDK 9

Char Array = String

but further not. } #

→ Random Access

Advantage

L arr[3]

→ caching